

PROGRAMME : Diploma Programme in Information Technology/Computer Technology

COURSE : Software Testing (STG)

COURSE CODE : 233118

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme						Credits	Assessment Scheme										Total Marks
Actual Contact Hrs./Week			SLH	NLH	Paper Duration		Theory				Based on LL and TSL				Based on Self Learning		
											Practical				SLA		
CL	TL	LL					FA-TH	SA-TH	Total	FA-PR		SA-PR		SLA			
03	-	02	01	06	03	03	Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	150
							30	70	100	40	25	10	--	--	25	10	

IKS Content: Total Learning Hours for Term: 00 Hrs

Indicates Online Examination, @ Internal assessment

1.0 RATIONALE:

This course covers all basic to advanced level concepts in software testing. It also includes technical as well as supporting skills necessary to become successful tester. In this course student will learn how to immediately find problems in any computer program, how to plan an effective test approach, how to clearly report your finding and how to tell when your software is ready for release.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The aim of the course is to help the student to attain the following industry /employer expected outcome: students will possess a foundational understanding of software testing principles and methodologies, enabling them to contribute effectively to quality assurance processes.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Identify impact of software bug and importance of software testing.
2. Select appropriate method for testing depending on purpose of testing and apply it.
3. Design and execute test cases for any software under test.
4. Execute test cases on software under test to validate its functionality.
5. Report the testing efforts in manual format and in defect tracking system and use various automation tools for testing.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I Software Testing Background	1a. Identify need of software testing. 1b. Define suspicious situation as a bug 1c. Describe various	1.1 Software error case studies – Disney Lion King, Intel Floating Point Division Bug, NASA Mars Polar Lander, Y2K Bug 1.2 Bug: Terms for software Failures, A Formal Definition, Bug occurrence, cost of bugs, Goal of software tester, Traits for Good software tester 1.3 Software Testing: Definition, goal	06

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	terminologies used in testing later on wherever needed.	of Software Tester 1.4 Testing Axioms 1.5 Software Testing LifeCycle 1.6 Software Testing Terms: Precision and Accuracy, Verification and Validation, Quality and Reliability, Quality Control Quality Assurance, V model.	
Unit-II Testing methodologies	2a. Distinguish between various testing methods. 2b. Identify purpose and accordingly select appropriate method. 2c. Apply various black box testing techniques.	2.1 Black Box and White Box Testing 2.2 Static and Dynamic Testing 2.3 Static Black Box Testing: Testing the Specification, Performing a High-level Review of the Specification, Low level Specification Test Techniques 2.4 Dynamic Black Box Testing: • Test-to-pass and Test-to-Fail, • Equivalence Partitioning, • Data Testing: Boundary Conditions, Sub-Boundary Conditions, Default, Empty, Blank, Null, Zero and None, Invalid, Wrong, Incorrect and Garbage Data • State Testing: Testing Software's Logic Flow, Testing States to Fail • Other Black Box Test Techniques: Behave like a Dumb User, Look for the Bugs where you have already found them, think like a Hacker, Follow experience, intuition and Hunches.	08
Unit-III Code Examination	3a. Distinguish between various testing methods. 3b. Identify purpose and accordingly select appropriate method. 3c. Apply various white box testing techniques.	3.1 Static White Box Testing: • Formal reviews: Peer Reviews, Walkthrough, Inspection • Coding Standards and Guidelines • Generic Code Review Checklist: Data Reference Errors, Data Declaration Errors, Computation Errors, Comparison Errors, Control Flow Errors, Subroutine Parameter Errors, Input/Output Errors 3.2 Dynamic White Box Testing • Testing the Pieces • Data Coverage: Data Flow, Sub	09

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		Boundaries, Formulas and Equations, Error Forcing Code Coverage.	
Unit-IV Types of testing and Test plan	4a. Differentiate among various types of testing. 4b. Identify appropriate type of testing according to requirement of testing. 4c. Test the software with respect to all types of testing. 4d. Test the website. 4e. Create Test plan.	4.1 Unit Testing: Stub and Driver 4.2 Integration Testing: Big Bang, Top-Down, Bottom-Up, Sandwich approach. 4.3 System Testing: Functional Requirement testing and Non Functional Requirement Testing. 4.4 Acceptance testing: Alpha and Beta testing. 4.5 Regression Testing: Regular regression and Final regression Testing. 4.6 Website Testing 4.7 Test Planning: The goal of the test Planning • Test Planning Topics: high level expectations, people, places and things, definitions, Inter group Responsibilities, what will and won't be tested, test phases, test strategy, resource requirements, tester assignments, test schedule, test cases, Metrics and Statistics, Risk and Issues.	10
Unit-V Test Documentation and Automation testing Tools	5a. Report the defects manually. 5b. Report the defect using defect tracking system. 5c. Differentiate between manual and automation Testing. 5d. Identify appropriate type of automation tool according to test requirement	5.1. Writing and Tracking Test Cases: Test case planning, test design, Test Case, test procedures, test case organization & tracking. 5.2. Test Report: A bug's life cycle, 5.3.1 Bug tracking system: Manual Bug Reporting and Tracking, Automated bug reporting and tracking. 5.3. The benefits of automation and tools. 5.4. Test tools: Viewers and Monitors, Drivers, Stubs, Stress and load tools, Interference injectors and noise generators, analysis tools. 5.5. Software Test Automation: Macro Recording and playback, programmed macros, Fully Programmable Automated Testing Tools. 5.6. Testing: Random Testing and Monkey tools. Realities of using test tools and automation.	12
		TOTAL	45

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
I	Software Testing Background	04	04	04	12
II	Testing methodologies	04	06	04	14
III	Code Examination	04	04	04	12
IV	Types of testing	04	08	04	16
V	Test Documentation and Automation testing tools	04	08	04	16
	TOTAL	20	30	20	70

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills (**Outcomes in cognitive, psychomotor and affective domain**) so that students are able to acquire the desired programme outcome/course outcome.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	Unit No.	Laboratory Learning Outcome	Laboratory Experiment Title	Hours
01	I	1a. Use Software tools to write problem statement and identify scope of the project.	Study any software system specification and design test cases.	4
02	II, IV	2a. Use appropriate process model and activities related to project.	Design test cases for Calculator application.	2
03	III, IV	3a. Apply the principles of requirement engineering.	Create any GUI application and report bugs.	4
04	IV	4a. Create SRS document for the project.	Perform testing of any website and report bugs.	2
05	IV	5a. Construct use case diagram for software models.	Design test cases for railway reservation form.	2
06	IV, V	6a. Design activity diagram for the project.	Design test cases for Social site (Twitter, Facebook) login form.	2
07	II, IV	7a. Draw data flow diagram for the project 7b. Create Decision tables and E-r diagram.	Write test cases for usability testing of website.	2
08	V	8a. Represent software project by Class Diagram.	Design and run test cases for WordPad using any automated tool.	4
09	V	9a. Prepare decision table for the Project.	Report the bugs using Bug Tracking Tool (e.g. JIRA).	2
10	V	10a. Identify risk involved in the Project.	Automate any application for test management tool (e. g. Test Manager).	4

Sr. No.	Unit No.	Laboratory Learning Outcome	Laboratory Experiment Title	Hours
		10b.Prepare RMMM plan.		
11	III,V	11a.Estimate size of the project using function point metric.	Perform unit testing on any s/w unit using automated tool(e. g. junit)	2
Total				30

7.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects.

1. Research and summarize: Unit, Integration, System, Acceptance, Regression, Smoke, and Sanity testing.
2. Explore any one tool (e.g., Selenium, JUnit, Postman, or Bugzilla) and write a short report on what you did.
3. Set up a basic automation framework (even a simple script) using a chosen tool (e.g., Selenium with Python/Java, Playwright with JavaScript/Python).
4. Use a tool like Postman to test a public API (e.g., a "dummy" REST API, weather API, or a simple CRUD API). Send GET, POST, PUT, and DELETE requests. Validate response status codes and body.
5. Research and understand the concepts of load testing, stress testing, and endurance testing.
6. Explore common web application security vulnerabilities (e.g., SQL injection, XSS, broken authentication).
7. Research the unique challenges and considerations for testing mobile applications (iOS/Android).
8. Find an existing open-source project or a website with known bugs. Select a bug and write a detailed bug report as if you were reporting it to the development team.
9. Choose one of the following and research it: Agile testing, Test-Driven Development (TDD), Behavior-Driven Development (BDD), or Shift-Left Testing.
10. Manual vs Automation Testing: Create a comparison table with advantages, disadvantages, and use cases.

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Guide students in undertaking mini projects.
2. Demonstrate students thoroughly before they start doing the practice.
3. Encourage students to refer different website and YouTube channels to have deeper understanding of the subject.
4. Observe and monitor the performance of students in lab.
5. Use of online platforms like MOOCs, Swayam, Spoken tutorial, NPTEL, Infosys springboard etc.
6. Arrange a visit to Software Industries for actually observing testing methodologies they are applying.
7. Arrange expert lecture of industry person in the area of Software testing and test automation.

9.0 ASSESSMENT METHODOLOGY

1. Formative assessment: TH
Periodic Test: 30marks
2. Summative assessment: TH
70 marks Final theory paper.
3. Formative assessment: PR
Lab performance, Viva
4. Self Learning assessment:

10.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Author	Title of Book	Publication
1	Software Testing	Ron Patton	PEARSON
2	Software Testing: Principles, Techniques and Tools	M. G. Limaye	Tata McGraw-Hill
3	Foundations of Software Testing	Rex Black, Dorothy Graham, and Erik van Veenendaal	Cengage Learning EMEA (4th edition)
4	Software Testing: A Craftsman's Approach	Paul C. Jorgensen	Auerbach Publications, June 29, 2021
5	The Art of Software Testing	Glenford J. Myers, Corey Sandler, and Tom Badgett	Wiley, November 2011

B) Software/Learning Websites:

1. http://en.wikipedia.org/wiki/Test_automation
2. http://en.wikipedia.org/wiki/Software_testing#Testing_tools
3. <https://www.softwaretestingmaterial.com>
4. <http://www.etestinghub.com/>
5. http://www.tutorialspoint.com/software_testing/
6. <https://www.softwaretestinghelp.com/manual-testing-tutorial-1/>

C) Major Equipment/ Instrument with Broad Specifications:

Sr.No.	Equipment	Specification
1	Desktop Computer	PC Specifications to be followed: Processor: i3 or i5 (or Higher) RAM: 4 GB (or Higher) HDD: 1 TB SATA Monitor: TFT LCD OS: Genuine Windows 8 or 10 Professional or Home Premium or Windows 8 or 10 Ultimate Internet Connection
2	LCD Projector	Display Type: LCD Light Output: 3200 Lumens
3	Software	Selenium, Junit, Jira, Test Manager

11.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)		
	1	2	3	4	5	6	7	1	2	3
CO1		H	M					L		
CO2		H	M	H	H	L		M	H	
CO3			L	H		H				H
CO4		M	L	H		H				H
CO5	H	M	H				M			M

H: High Relationship, M: Moderate Relationship, L: Low Relationship.